3.2.9 I: Differential equations

	Content
I1	Find and use an integrating factor to solve differential equations of the form $\frac{\mathrm{d}y}{\mathrm{d}x} + \mathrm{P}(x)y = \mathrm{Q}(x)$ and recognise when it is appropriate to do so.

	Content
I2	Find both general and particular solutions of differential equations.

	Content
I3	Use differential equations in modelling in kinematics and in other contexts.

	Content
I4	Solve differential equations of the form $y'' + ay' + by = 0$ where a and b are constants, by using the auxiliary equation.

	Content
I5	Solve differential equations of the form $y'' + ay' + by = \mathrm{f}(x)$ where a and b are constants by solving the homogeneous case and adding a particular integral to the complementary function (in cases where $\mathrm{f}(x)$ is a polynomial, exponential or trigonometric function).

	Content
I6	Understand and use the relationship between the cases when the discriminant of the auxiliary equation is positive, zero and negative and the form of solution of the differential equation.

	Content
I7	Solve the equation for simple harmonic motion $\ddot{x} = -\omega^2 x$ and relate the solution to the motion.

I8	Model damped oscillations using 2nd order differential equations and interpret their solutions. Understand light, critical and heavy damping and be able to determine when each will occur.

	Content
I9	Analyse and interpret models of situations with one independent variable and two dependent variables as a pair of coupled 1st order simultaneous equations and be able to solve them, for example predator-prey models.

	Content
I10	Use of Hooke's Law with $T = kx$ to formulate a differential equation for simple harmonic motion, where k is a constant.

	Content
I11	Use models for damped motion where the damping force is proportional to the velocity.